Stereo 3D / 6DoF Manager

Complete Technical Handoff

Build	86fc41b8
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Platform	Electron 31 desktop application, Windows x64
Managed mods	17
Cached core sources	17
Game database	3,335 titles
IPC surface	99 handlers, 99 bridged
Automated checks	71,000+ across 21 suites

This document is the complete technical description of the application: what it manages, where every file comes from, how each mod is installed and configured, how the scanner identifies a game, how downloads survive a hostile network, and what is verified automatically. It is written to be sufficient for someone who has never seen the codebase to take ownership of it.

Honesty note. Where something is unverified, partially implemented or known to be fragile, this document says so explicitly rather than omitting it. Section 27 is a complete ledger of known gaps, and section 26 lists every significant bug found during development with its root cause. A handoff document that only lists successes is not useful to whoever inherits the work.

Contents

1. What the application does

The application finds installed PC games, works out which stereoscopic 3D and 6DOF head-tracking mods can run on each one, downloads those mods from their original sources, installs them into the correct folder, and exposes their real configuration files for editing. It removes them cleanly when asked.

The problem it solves

Stereoscopic 3D modding on PC is fragmented. The drivers live in different places — a blogspot page, several GitHub repositories, a personal website — and each has its own packaging, its own installation rules, and its own idea of where files belong. A driver that works on DirectX 11 will do nothing on DirectX 12. Two drivers that both hook `d3d11.dll` cannot coexist. A mod built for one game will not work on another. Getting a single game running stereoscopically can mean an hour of reading forum threads.

The application encodes that knowledge. It knows which driver suits which rendering API, which files each package contains and where they belong relative to the game executable, which mods contend for the same DLL entry point, and how to resolve that contention when a resolution exists.

What it explicitly does not do

- **It does not host or redistribute any mod.** Every file is fetched from the author's own release page or website at install time. Nothing is mirrored, repackaged or bundled, with one documented exception.
- **It does not modify game files.** Mods are additive: DLLs, shaders and configuration files placed alongside the game executable. Uninstalling removes exactly what was placed and nothing else.
- **It does not bypass any protection.** Games that refuse to run with a proxy DLL present, typically because of anti-cheat, are outside its scope.
- **It does not encourage use in multiplayer.** Injecting a DLL into a game with anti-cheat can result in a ban. The interface says so at the point of installation.

Scope in numbers

Quantity	Value	Note
Managed mods	17	Drivers, add-ons, hosts and loaders
Core download sources	17	Cached once, linked into many games
Game database	3,335	Name, engine, API and bitness hints
Rendering APIs	8	DX7, DX8, DX9, DX10, DX11, DX12, Vulkan, OpenGL
Output formats	47	Across the seven mods offering a choice
Engine layouts	11	Unreal 4/5, Unity, RedEngine, CryEngine, Source, RE Engine and others
Executable shapes tested	103	Real folder structures from two decades of releases
6DOF mods reachable	80	50 from the BerZerker hub, 30 from itsloopyo

2. Architecture and process model

A standard two-process Electron application. All privileged work — filesystem, network, process spawning — happens in the main process. The renderer is a single HTML document with no Node integration, reaching the main process only through an explicitly enumerated preload bridge.

Process responsibilities

Process	Owns	Never does
Main	Filesystem, HTTPS, archive extraction, PE parsing, process launch, logging, settings	Renders UI or assumes anything about the DOM
Preload	A fixed list of 99 channel bindings; context isolation on	Expose `require`, `fs` or arbitrary IPC
Renderer	All presentation, all user interaction, all state display	Touch the filesystem or the network directly

The bridge is exact: 99 handlers registered in the main process, 99 invocations exposed in preload. That symmetry is asserted by the test suite, so a handler added without a binding — or a binding pointing at a handler that does not exist — fails the build rather than surfacing as a silent no-op at runtime.

Why the renderer has no Node access

The renderer displays data derived from arbitrary files on disk and arbitrary HTML fetched from the internet: release pages, asset names, game folder names. Any of those could contain hostile content. With context isolation on and no Node integration, the worst a malicious asset name can do is display incorrectly.

Everything crosses one logged boundary

Every IPC call is wrapped once, centrally, so each user action is recorded with its arguments, duration and result. This was added after several rounds of debugging where the question was not “what went wrong” but “did the click even arrive”. Arguments are summarised rather than dumped — long strings truncated, large arrays counted — and anything matching `token`, `password`, `secret` or `authorization` is written as `[redacted]`, so a log is safe to attach to a bug report.

```
[IPC ->] htInstallManual  {"args":[{"source":"itsloopyo","game":"Peak", ...}]}
[IPC <-] htInstallManual  {"ms":11840,"result":{"ok":true,"tag":"v1.1.2"}}
```

Failure isolation

Uncaught exceptions and unhandled promise rejections in the main process are captured and written to the log rather than terminating silently. Every install is wrapped so that a filesystem error becomes a described result rather than an exception crossing the IPC boundary, where it would surface to the user as a bare errno string.

3. The file layout

Source modules

File	Size	Responsibility
main.js	~1,200 lines	Window lifecycle, 99 IPC handlers, icon resolution, process launch
preload.js	~130 lines	The context bridge — the complete list of what the UI may ask for
renderer/index.html	~407 KB	The entire interface: markup, styling and logic in one document
src/installer.js	4,013 lines	Downloading, extraction, placement, configuration, conflict resolution, uninstall
src/scanner.js	728 lines	Drive enumeration, game discovery, executable identification, API and bitness detection
src/mods.js	~450 lines	The catalogue: every mod, its source, placement rules and API support
src/gamedb.js + ext	~4,000 lines	3,335 known titles with engine, API and bitness hints
src/ghfree.js	~150 lines	GitHub access that does not depend on the rate-limited API
src/peicon.js	~200 lines	PE resource parsing, to read an icon out of an executable
src/logger.js	~150 lines	Two rotating logs beside the executable
src/store.js	~200 lines	Settings, profiles and library persistence
src/config.js	~250 lines	INI, XML and ReShade preset reading and writing

Runtime locations

Path	Contents	Survives update
%APPDATA%\Stereo3D Manager\	settings.json, library.json, profiles.json	Yes — outside the program folder
…\core\	Cached mod payloads, one folder per source and version	Yes
…\cache\github\	Cached release metadata, ten-minute TTL	Yes
<app>\logs\	app.log, download.log, update.log	Yes — excluded from the update copy
<app>\manual-core\	Manually supplied mod packages	Yes — excluded from the update copy
<game>\.stereoscope\manifest.json	What this application installed into that game	Lives with the game

Why settings moved. They were originally written next to the executable, which is precisely the folder an update replaces — so updating the application destroyed the user's library. They now live in the per-user data folder alongside the mod cache, so a settings file and the mods it refers to stay together. Existing installations are migrated once, automatically, and the originals are kept with a `.migrated` suffix rather than deleted.

4. Data flow: scan to install

Scanning

```
drive enumeration            A-Z, readdir probe, EPERM/EBUSY counts as present
  -> Steam libraries         read explicitly from libraryfolders.vdf
  -> general walk            depth 7, junk folders excluded
  -> rank executables        every candidate in the tree, scored against each other
  -> read the binary         render API and bitness from its imports
  -> match the database      folder name against 3,335 known titles
  -> merge                   keeping games already known on other drives
```

A per-drive scan runs the same three discovery steps as a full scan and filters the result to that drive. That symmetry matters: an earlier implementation used a different method for per-drive and returned zero games on Steam drives, because it skipped the explicit Steam library enumeration that a full scan performs first.

Installing

```
resolve the source          catalogue entry, or the exact build the user picked
  -> ensure the core        manual-core, then disk cache, then download
  -> verify the archive     magic bytes: PK, 7z, Rar!
  -> extract                zip / 7z / rar; wrapper folders unwrapped
  -> check the slot         refuse or rehome on proxy contention
  -> preflight              writable? at least 64 MB free?
  -> place                  per-mod rules, relative to the executable
  -> verify each write      size check, to detect antivirus quarantine
  -> seed configuration     defaults, without overwriting user values
  -> record the manifest    exact file list, for a clean uninstall
```

Every step can fail with a specific, actionable message rather than a generic error. The complete set of failure modes is enumerated in section 19, and each one is covered by a test.

Uninstalling

The manifest is read, its file list is removed, and the entry is deleted. Nothing is guessed at. Where a mod was rehomed — renamed to a different proxy slot to make room for another — a sidecar record ensures the renamed file is removed too.

5. The scanner: finding the real executable

This is the hardest problem in the application and the one that took the most iterations to get right. Everything downstream depends on it: the rendering API is read from the chosen binary, mods are installed beside it, and the launch button runs it.

Why it is hard

A game folder rarely contains one executable. It contains the game, plus some combination of launcher, crash reporter, anti-cheat service, redistributable installers, shader compiler, save-editing tool, texture converter, mod manager and video player. Several of those are larger than the game. Several link the same graphics APIs the game does. Some resemble the folder name more closely than the game does.

Worse, the game is frequently not in that folder at all. It might be in `Binaries\Win64`, `bin\x64`, `runtime\media`, `Game\`, or three levels down a path nobody has documented.

Real examples that broke naive approaches

Game	Correct executable	What a naive scan picked
Final Fantasy VII Remake	End\Binaries\Win64\End-Win64-Shipping.exe	Launcher.exe at the root
Star Wars Jedi: Survivor	SwGame\Binaries\Win64\JediSurvivor-…	SwGame.exe at the root
Judgment	runtime\media\Judgment.exe	nothing — folder shape unknown
Persona 5 Royal	P5R.exe	bin\SaveRemoverTool.exe
Metaphor: ReFantazio	METAPHOR.exe	bin\crash_handler.exe
Baldur's Gate 3	bin\bg3.exe	bin\nvdxt.exe, CefSharp…exe
Dark Souls III	Game\DarkSoulsIII.exe	Game\DS3T.exe (mod tool)
Dishonored	Binaries\Win32\Dishonored.exe	leaveingamedir.exe
The Last of Us Part I	tlou-i.exe	crs-video.exe (movie player)
Trails in the Sky FC	ed6_win_DX9.exe	_Redist\QuickSFV.EXE
VaM	VaM.exe	installer_files\miniconda_installer.exe (80 MB)
Skyrim Special Edition	SkyrimSE.exe	CreationKit.exe
Skyrim VR	SkyrimVR.exe	hkxcmd.exe (Havok tool)
Kingdom Come II	Bin\Win64MasterMasterSteamPGO\KingdomCome.exe	Update\Patch.exe
Persona 4 Golden	P4G.exe	…Plus 29 Trainer Updated.exe
Immortals of Aveum	x64\Windows\…	listed as a game called "Windows"

Every row came from a real 117-game library, not from imagination.

The approach that works

Three signals combined in strict precedence, with a hard exclusion layer underneath.

- **Exclusion first.** Folders that never contain a game are not walked at all. This is what defeats the 80 MB miniconda installer: it never becomes a candidate, so its size cannot help it win.

- **Then rank every executable in the tree against every other.** Not the first match in a list of known folder names — that approach returned `bin\SaveRemoverTool.exe` because `bin` was on the list and the function returned early, before anything looked at `P5R.exe` in the root.
- **Named folder shapes are a hint, not an answer.** A candidate found in a known-good folder wins ties, because a named shape is meaningful evidence — but it cannot override a clearly better binary elsewhere.

Named shapes recognised

The hint list covers the layouts that are known and stable. It is a fast path, not the authority.

```
Unreal          <Project>\Binaries\Win64 | Win32 | WinGDK      Engine\Binaries\Win64
RGG / Dragon    runtime\media
FromSoftware    Game\
Square Enix     game\                                         End\Binaries\Win64
REDengine       bin\x64        bin\x64_dx12
CryEngine       Bin64\         bin\win_x64
Source          game\bin\win64
Firaxis         Base\Binaries\Win64Steam
Blizzard        _retail_\
Legacy          System\        (Unreal 1/2 era)
Generic         bin\  Bin\  x64\  bin64\  Binaries\  retail\  client\  app\
```

6. Executable ranking in detail

The score

Signal	Weight	Rationale
`-Win64-Shipping.exe`	+100	Unreal always names the real binary this way; near-conclusive
`-WinGDK / -Win32-Shipping`	+80	The same, for Game Pass and 32-bit builds
Generic `-Shipping.exe`	+70	Older Unreal projects
Links a graphics API	+25 × tier	75 for D3D12/Vulkan/DXGI/D3D11, 50 for D3D9/D3D8/DDraw/OpenGL, 25 windowing-only
Size band	+40 / +30 / +20 / +8	Thresholds at 40 MB, 10 MB, 3 MB, 512 KB
Name resembles the folder	+35	“witcher3” inside “The Witcher 3”
In a binaries-like folder	+15	Win64, bin, Binaries, retail, media
DX12 build of a pair	+12	Control_DX12 over Control_DX11
64-bit twin	+6	PathOfExile_x64 over PathOfExile
Depth	–3 per level	Prefer the shallower of two equals
`Engine\` fallback build	–10	UE4Game-Win64-Shipping loses to a project’s own build
Looks like a tool	–**200**	A penalty, not an exclusion — see below

Why the graphics test matters more than the name

Name blocklists are endless. Every new library turns up another `pygrun.exe` or `LayersChecker.exe`. The binary itself carries a far stronger signal: a game links Direct3D, OpenGL or Vulkan; a save editor, packer or crash reporter does not. Reading the import strings requires no list to maintain and generalises to tools nobody has seen yet.

```
TheGame.exe          3        SaveRemoverTool.exe    0
OldGame.exe          2        pygrun.exe             0
                              BugSplatHD64.exe       0
```

The result is cached per file and reads only the first six megabytes, where the import table lives, so the cost stays bounded across a large library.

Why size is banded rather than compared directly

A game executable is usually far larger than the tools beside it, so size is a strong signal. But comparing raw bytes would let a 4.1 MB tool edge out a 4.0 MB game. Banding keeps them tied and lets a better signal decide. Raw byte comparison remains only as the final tie-break.

Verified precedence

Scenario	Result	Why it matters
60 MB game vs 2 MB editor, both link D3D11	Game	The graphics test alone cannot separate them
600 KB game vs 80 MB installer	Game	Size alone would get this wrong
4.0 MB DX12 vs 4.1 MB DX11	DX12	Banding stops raw bytes deciding

Scenario	Result	Why it matters
SkyrimSE 60 MB vs CreationKit 30 MB, both render	SkyrimSE	The case neither signal solves alone
Small binary vs five large launchers	The binary	Tool penalty outranks size entirely

Penalty, not exclusion

A name that looks like a tool subtracts two hundred points rather than removing the candidate. This matters for a folder containing nothing but oddly-named binaries: the scan still returns the most game-like one instead of nothing at all. Verified with a folder holding only `converter.exe` and `packer.exe` — it answers rather than failing.

What counts as a tool

Two tiers. **Never** means skipped outright; **demote** means sorted last but still eligible. Both were built by category from real installations, and every entry below is asserted in the test suite.

Category	Examples
Unreal tooling	ShaderCompileWorker, UnrealPak, UnrealLightmass, SwarmAgent, UE4PrereqSetup, CrashReportClient
Epic / EOS	EpicWebHelper, EOSBootstrapper, EpicGamesLauncher, UnrealCEFSubProcess
Anti-cheat	EasyAntiCheat, BEService, BEClient, PunkBuster, GameGuard, npkcrypt, Xigncode, xhunter
DRM	Denuvo, VMProtect, Themida
Launchers	Uplay, Ubisoft Connect, GalaxyClient, Origin, EA Desktop, Rockstar, Battle.net, updaters, patchers
Runtimes	PhysX, XNA, .NET, DirectX redist, OpenAL, vcredist
Middleware	Wwise, FMOD setup, Havok, hkxcmd, SpeedTree, Coherent, Awesomium
Crash reporting	WerFault, breakpad, crashpad_handler, Sentry, BugSplat, CrashSender
Overlays	RTSS, Afterburner, ShadowPlay, Overwolf, Fraps, Bandicam
Media players	crs-video, bink, Smacker, CriWare, ADX2, Scaleform
Mod tooling	3DMigoto Loader, nvdxt, texconv, CreationKit, xEdit, LOOT, Wrye, MO2, Vortex, NifSkope, trainers
Legacy cruft	dxdiag, regsvr32, AcroRead, install/readme/manual executables, uninstallers

The half that matters more. A broad blocklist is worthless if it eats real games. The suite asserts 115 shipping executables are never demoted — spanning Unreal 2 through 5, RE Engine, RAGE, REDengine, Creation, id Tech, Decima, Source, Unity, Godot, RGG and FromSoftware, from `UT2004.exe` and `DeusEx.exe` to `ff7rebirth_.exe` and `tlou-ii.exe`. One pattern, `^cri`, was caught this way: it matched **Cri**msonDesert.

7. Folders never searched

Two mis-picks came from this layer rather than from ranking, and no filename rule could have caught either. Nothing inside a redistributable or installer folder is ever the game, so the fix is to not look there at all.

The two cases

```
Trails in the Sky\_Redist\QuickSFV.EXE
VaM\TextAudioTool-0.4\installer_files\miniconda_installer.exe   (80 MB)
```

A checksum utility called QuickSFV would defeat any name heuristic — the name says nothing. And the miniconda installer is more than five times the size of VaM itself, so size scoring actively favoured it. Excluding the folder solves both without a single filename rule.

The exclusion list

Category	Folders
Redistributables	_Redist, _CommonRedist, redistributable, vcredist, DirectX, dotnet, prerequisites, prereq
Installers	installer_files, installers, __Installer, dxsetup, _MEI*
Development	tools, sdk, modtools, editor, devkit
Documentation	docs, manual, support, extras, soundtrack, artbook
Runtime clutter	logs, dumps, crashes, screenshots, savegames, backup, _backup, cache, temp
System	$*, Windows, Windows.old, System Volume Information, Recovery, ProgramData, AppData, node_modules

Asserted in both directions. The suite checks that each of these is excluded *and* that `Binaries`, `Win64`, `bin`, `Game`, `runtime`, `media`, `Content` and `Engine` are still walked. Widening an exclusion list is exactly the kind of change that quietly breaks real detection.

Binary folder matching is by prefix

Real builds append configuration names to the platform folder. Kingdom Come ships `bin\Win64MasterMasterSteamPGO`; others use `Win64Shipping` or `Win64Retail`. Exact matching missed all of them.

8. Game naming and library hygiene

Deriving a name

A folder name is only the game's name if it is not a platform or plumbing folder. `Immortals of Aveum\x64\Windows` was being listed as a game called "Windows". The resolver walks up past plumbing folders — `x64`, `Windows`, `bin`, `Game`, `Update`, `DotNetCore`, `Win64*` — while refusing to climb into a library root such as `common\` or `steamapps\`, which would name every game after its store.

Path	Resolved name
G:\Immortals of Aveum\x64\Windows	Immortals of Aveum
E:\…\common\Baldurs Gate 3\bin	Baldurs Gate 3
E:\…\KingdomComeDeliverance2\Bin\Win64MasterMasterSteamPGO	KingdomComeDeliverance2
E:\Dark Souls 3\Game	Dark Souls 3
E:\…\common\StellarBlade	StellarBlade

It is also separator-agnostic: a Windows path resolves identically wherever the code runs, which matters because the test suite runs on Linux.

Nested entries

A game folder contains `bin\`, `Launcher\`, `Update\` and similar, and each can look like a game on its own. Baldur's Gate 3 appeared four times in one real library — once correctly, and three times as subfolders holding stray tools. Only the outermost folder is kept; nine candidates collapse to three games.

A scan merges; it does not rebuild. This was a subtle trap that cost two rounds of debugging. Fixing the scanner did not fix libraries that already contained bad rows, because those rows were never re-derived — a scan adds and updates, it does not discard what it did not find. Nested entries are therefore also pruned when the library loads, once, with the cleaned result saved. Manually added games are never pruned.

An empty scan cannot clear the library

The original code emptied the games array before examining the scan result, so a scan that found nothing destroyed the entire library. It now returns early with a message naming the drive: "No games found on E:\ — your library is unchanged".

Per-drive merging

A per-drive scan replaces only the games on that drive. An earlier version cleared everything and repopulated from one drive, so scanning G: silently deleted every game found on C: and D:.

Exclusions

Folders can be excluded permanently from Settings. This is the correct answer for cases detection cannot reasonably solve — an emulator sitting in a folder named after a game, a VR utility, a toolkit that ships as its own Steam app.

9. Icon resolution

Six strategies in descending order of quality. The chain exists because a shipping binary buried in Binaries\Win64 frequently carries no icon resource at all, and the Windows shell returns nothing for it.

#	Strategy	Notes
1	Steam library artwork	Best quality when the game is a Steam title
2	An .ico or .png beside the executable	Common for indie and GOG releases
3	Artwork in the game root	Walks up four levels for icon.ico, logo.png and similar
4	**PE resource extraction**	Reads the icon out of the executable directly
5	The Windows shell	`app.getFileIcon`, as a fallback
6	A sibling or launcher executable	A launcher usually carries the game's artwork

PE resource extraction

The reliable path for non-Steam games. The parser reads the DOS header to find the PE header, takes data-directory entry 2 for the resource table, maps virtual addresses to file offsets through the section table, then walks the three-level resource tree — type 14 (RT_GROUP_ICON) for the directory of sizes, type 3 (RT_ICON) for each image — and reassembles a valid .ico. Where a frame is stored as PNG, as it is in Vista-era and later icons, that is returned directly.

```
dgVoodooCpl.exe -> 4 frames: 256x256, 48x48, 32x32, 16x16
                   ico = 63,735 bytes,  png = 56,313 bytes
PNG signature: 89504e470d0a1a0a      dimensions: 256x256
```

Two things this buys beyond simply working. It picks the *largest* icon present, where the shell often returns 32×32 even when a 256×256 exists — which is why some icons looked poor rather than missing. And it runs before the shell, asserted by comparing source positions so a future edit cannot silently demote it. An executable with no icon resource returns null cleanly rather than throwing.

The Steam artwork cache

A bug worth recording. The code derived the artwork cache path from the game's own location, so a game in `D:\SteamLibrary\` looked for `D:\SteamLibrary\appcache\librarycache`, which does not exist — Steam keeps that cache only in its install root. Every game outside the drive Steam is installed on therefore had no artwork, which presented as "some games have no icon" with no obvious pattern. The resolver now collects every plausible Steam root including the registry's `SteamPath` value.

Downloading missing artwork

For games that genuinely have no icon anywhere on disk, a Settings action resolves the name through Steam's public app-search endpoint and fetches store artwork from the CDN, preferring the tall library capsule, then the wide capsule, then the header. No key or login is required — these are the same URLs a store page uses. It only touches games still showing a placeholder, validates the bytes are a real JPEG before accepting them, shows progress, and marks failures so an unresolvable name is not retried on every run.

Ambient glow

The area behind each icon is tinted with a colour sampled from the artwork. The sampler draws the icon to an 18×18 offscreen canvas and averages it, ignoring transparent padding, skipping near-black and near-white pixels, and weighting saturated pixels about seven times higher. Without that weighting most game covers average to grey. Hue is preserved while saturation and lightness are pushed into a usable band, so no card ends up glaring or washed out. Results are cached per icon URL, and the whole effect can be switched off.

10. The mod catalogue

Seventeen managed mods across four roles: stereo drivers that produce the 3D image, VR export add-ons that send it to a headset, host frameworks the drivers run inside, and head-tracking mods that add positional camera control.

Mod	Role	Source	Config file
wiz3D (iZ3D wrapper)	host	effcol/wiz3D	wiz3D_Config.xml
3DVision4All (oneup03)	host	oneup03/3DVision4All	3dvision4all.ini
ReShade (add-on build)	host	reshade.me/	ReShade.ini
geo-11	mod	helixmod.blogspot.com/search/la…	d3dxdm.ini
geo-11 (GitHub mirror)	mod	ThreeDeeJay/geo-11	d3dxdm.ini
SuperDepth3D	mod	BlueSkyDefender/Depth3D	ReShadePreset.ini
Geo3D	mod	Flugan/Geo3D	ReShade.ini
Legacy Geo3D	mod	—	ReShade.ini
SuperVRExport add-on	addon	BerZerker96/Super-VRExport-Addon	—
GeoVRExport add-on	addon	BerZerker96/Super-VRExport-Addon	—
dgVoodoo2 (DX8/9/10 → DX11)	tool	dege-diosg/dgVoodoo2	dgVoodoo.conf
Head-Tracking (itsloopyo)	mod	—	—
6DOF Hub (BerZerker96)	mod	BerZerker96/6DOF-Head-Tracking-…	—
Osiris VR Viewer	tool	BerZerker96/Osiris-Vr-Viewer	—
BepInEx 5 (Unity loader)	loader	github.com/BepInEx/BepInEx/rele…	—
Ultimate ASI Loader	loader	github.com/ThirteenAG/Ultimate-…	—
REFramework (RE Engine)	loader	github.com/praydog/REFramework-…	—

How a core is resolved

Strategy	Used by	Mechanism
github-release	Most mods	Release feed, newest release regardless of pre-release flag
github-repo	SuperDepth3D	A branch archive, tracked by commit
website	ReShade	Site scrape, then a probe, then a pinned known-good version
blog-tiered	geo-11 official	Blog feed, then an S3 listing, then a probe, then pinned
bundled	Geo3D stable	Ships with the application; no download
per-game	6DOF mods	Downloaded per game rather than cached as a shared core

Provenance

Every entry is downloaded from its author's own release feed or website at install time. The application caches what it downloads so that ten games sharing a driver cost one download, but it never republishes anything. A manual-core folder lets a user supply a package by hand, which is also what makes the download path testable offline.

The one exception. A bundled build of Geo3D ships with the application, labelled “stable (bundled)” in the interface, because that specific build is no longer downloadable from its original location. It is presented as a distinct catalogue entry alongside the live download so a user always knows which they are installing.

11. Stereo 3D drivers, one by one

SuperDepth3D

Property	Value
Rendering APIs	DX7, DX8, DX9, DX10, DX11, DX12, Vulkan, OpenGL
Configuration file	ReShadePreset.ini
Source	BlueSkyDefender/Depth3D
Output formats	vr_addon, sbs, tab, interlaced, column_interlaced, checkerboard

A depth-buffer shader by BlueSkyDefender that runs inside ReShade. It reconstructs a second eye view from the depth buffer the game already produces, which is why it works on every rendering API rather than one specific one. It is the safety net: if nothing else covers a game's API, this will.

The trade-off is that a reconstructed view is not true geometry. Objects can show haloing at depth discontinuities, and transparent surfaces — water, glass, particle effects — often sit at the wrong depth because they are not written to the depth buffer. For a DX11 title geo-11 will look better; for a Vulkan title this is the only option.

Placement

```
reshade-shaders\Shaders\SuperDepth3D.fx      (flat, beside the executable)
reshade-shaders\Shaders\*.fxh                 (support headers)
ReShadePreset.ini  ->  [SuperDepth3D.fx]
```

Configuration

The shader technique is enabled in the preset at install time so it is active on first launch rather than requiring the user to find it in the ReShade overlay. Depth adjustment, separation and the depth-buffer selection are exposed through the configuration editor.

A real bug worth recording. The GitHub archive wraps everything in `Depth3D-master/`. The placement rule looked for a folder named `Shaders` at the top level, did not find it, and fell back to copying the entire repository — producing `reshade-shaders\Shaders\Depth3D-master\Shaders\SuperDepth3D.fx`, a path ReShade does not scan. The mod appeared installed and did nothing. The resolver now searches for the named folder wherever it actually is.

geo-11 (HelixMod, official)

Property	Value
Rendering APIs	DX11
Configuration file	d3dxdm.ini
Source	helixmod.blogspot.com/search/label/geo-11
Output formats	katanga_vr, sbs, tab, interlaced, checkerboard, nvidia_dx11, nvidia_dx9

A geometry-based stereo driver derived from 3Dmigoto. Rather than reconstructing a second view, it renders the scene twice with a shifted camera, producing genuinely correct stereo geometry. This is the best image quality available and the reason to prefer it whenever the game is DirectX 11.

The cost is scope and effort. It is DX11 only, and many games need a per-game shader fix from the HelixMod community to correct effects that break under a shifted camera — shadows, screen-space reflections, certain post-processing passes. The application links the HelixMod game list directly so those fixes are one click away.

Placement

```
d3d11.dll        (the proxy the game loads)
d3dxdm.ini       (stereo settings)
d3dx.ini         (3Dmigoto base configuration)
  chosen from the 32\ or 64\ tree by the game's bitness
```

Configuration

Separation and convergence live in `[Stereo]` inside `d3dxdm.ini`; loader, logging and rendering sections live in `d3dx.ini`. The editor resolves each section to the correct file automatically, so the user does not need to know which setting lives where.

Discovery is tiered. The official build is distributed through a blogspot post rather than a release feed, so the resolver tries the blog feed, then an S3 listing, then a direct probe, then a pinned known-good version. A GitHub mirror is offered as a separate catalogue entry so the user can choose deliberately rather than silently receiving a different build.

Geo3D (Flugan)

Property	Value
Rendering APIs	DX9, DX10, DX11, DX12
Configuration file	ReShade.ini
Source	Flugan/Geo3D
Output formats	vr_addon, sbs, tab, interlaced, column_interlaced, checkerboard, anaglyph

Geometric stereo delivered as a ReShade add-on rather than a proxy DLL. Because it runs inside ReShade it composes with other ReShade effects, and because it is geometric it avoids SuperDepth3D's transparency problems.

Two builds are offered. The live download tracks development and is marked unstable in the interface. A bundled stable build ships with the application for the reason described in section 10.

Placement

```
Geo3D.addon64 / Geo3D.addon32     (ReShade add-on)
3DToElse.fx                       (output conversion shader)
ReShade.ini  ->  [Geo3D]
```

Configuration

Geo3D writes its `[Geo3D]` section into `ReShade.ini` at runtime. The installer seeds that section with defaults so settings are editable *before* the first launch, which is otherwise a chicken-and-egg problem: the file does not exist until the game has run once. The `3DToElse` technique is enabled and configured for frame-sequential input and side-by-side output.

wiz3D

Property	Value
Rendering APIs	DX7, DX8, DX9, DX10, DX11, OpenGL
Configuration file	wiz3D_Config.xml
Source	effcol/wiz3D
Output formats	interlaced, sbs, checkerboard, anaglyph, shutter, sr_weave

An iZ3D-derived wrapper with the widest API reach of any driver here. The shipped package contains separate trees for DX8, DX9, DX10/11, OpenGL quad-buffer, AMD HD3D and 3D Vision Direct; the installer copies the tree matching the game's API and bitness.

It is a wrapper rather than an add-on, so it claims a DirectX entry point and cannot coexist with another mod wanting the same one. It also offers the widest set of output formats, including anaglyph and active shutter, which makes it the practical choice for unusual display hardware.

Placement

```
d3d9.dll | d3d11.dll | d3d8.dll | opengl32.dll
wiz3D_Config.xml
OutputMethods\*.dll
  selected from dx8\, dx9\, dx10-11\, opengl-quad-buffer-stereo\,
  hd3d\ or 3d-vision-direct\, then by x86 or x64
```

Configuration

Settings are stored as XML attributes — `<Separation Value="0.5"/>` — which the configuration reader parses natively rather than through a generic INI path. The file ships under three different names depending on the build (plain, HD3D, 3D Vision Direct); the resolver finds whichever is present, so an edit never lands in a file the mod does not read.

Hotkeys were corrected from the shipped configuration header, not from documentation. The guide previously stated that convergence had no hotkey and lived only in the XML. It does have one. Toggle 3D is Numpad *, separation is Numpad +/–, convergence is Shift+Numpad +/–, auto-focus is Numpad /, presets are Numpad 7/8/9, eye swap is Ctrl+F8, the on-screen display is Shift+F1.

3DVision4All

Property	Value
Rendering APIs	DX7, DX8, DX9, DX10
Configuration file	3dvision4all.ini
Source	oneup03/3DVision4All
Output formats	sbs, tab, row_interlaced, column_interlaced, checkerboard, leiasr, katanga

Revives NVIDIA 3D Vision on modern drivers by proxying `nvapi`. Useful for older titles that shipped with first-party 3D Vision support, and for display hardware that still speaks the 3D Vision protocol.

It ships four interchangeable proxy names — `dinput8`, `winmm`, `version` and `dsound` — which is what makes proxy rehoming possible when another mod wants the same slot. Only one is installed; the others exist as alternatives.

Placement

```
dinput8.dll          (primary; alternates used when the slot is taken)
3dvision4all.ini
EnableWindowed3D.exe (run once, elevated, per game folder)
```

Configuration

Sections are `[stereo]`, `[render]` and `[debug]`. Output mode, eye swap, forced windowed mode, DirectFlip defeat and render dimensions are all exposed. The elevated helper is run once per game folder after installation.

12. 6DOF head tracking

Two independent sources, covering different games. Both listen to head-tracking data over UDP — typically from opentrack — and move the in-game camera with six degrees of freedom rather than the two a mouse provides.

BerZerker96 hub

A single repository holding per-game mods as separate releases: roughly fifty releases, each a different game. Most are plain 6DOF. A subset are **combined 3D + 6DOF builds** that include their own stereo implementation and therefore need no separate stereo driver at all.

The combined builds are structurally different

	Plain 6DOF	Combined 3D + 6DOF
Loader	HeadTracking.asi + bundled ASI loader	**version.dll proxy**
Config file	HeadTracking.ini	**"3D-6DOF config.ini"** (with a space)
Sections	Pose, FOV, Network, Position, General	+ Stereo3D, Rotation, ZAxis, CullGuard, Advanced
Stereo	none	**its own DX11 implementation**
Needs ReShade	no	no — explicitly not

A bug this exposed. The configuration editor filtered to a declared list of section names that did not include `Stereo3D`. A combined build would install and run correctly, but every 3D setting it has — separation, convergence, output mode, eye swap and all the stereo hotkeys, twenty-eight keys in total — was invisible in the interface. The mod's headline feature was unconfigurable. The editor also only looked for `HeadTracking.ini` and never the combined build's filename.

Title and tag frequently differ

A release's display title and its git tag are often not the same. `ACOrigins` is tagged `1`, `ACSyndicate` is tagged `s1`, `ACUnity` is tagged `u1`. Requesting a release by title returns 404 and the version list came back empty. The resolver matches on either `tag_name` or `name`, and tries the picked tag before falling back to the game name.

itsloopyo

One repository per game, around thirty of them. Roughly half publish only pre-releases.

Two bugs specific to this source

- **Pre-releases were invisible.** GitHub's `/releases/latest` endpoint excludes pre-releases and returns 404 when a repository has only those. Fourteen of thirty mods appeared to have no downloads at all. The resolver now lists all releases and takes the newest regardless of the flag.
- **Asset filtering excluded the only available asset.** Released mods ship a `-nexus.zip`; pre-release mods have no Nexus page and ship only `-installer.zip`. The rule excluded installer archives, so pre-release mods resolved to no asset and the download failed. Asset selection now filters only GitHub's auto-generated source archives.

Both sources are always available

Neither section is ever greyed out. Even when the catalogue has no entry matching a game's name, the manual picker can install any mod from either source onto any game — so "Unavailable" was actively misleading. The button stays live and reads "Pick a mod below", with a tooltip explaining why. BerZerker is framed red and itsloopyo blue so the two blocks are distinguishable at a glance when stacked.

Manual installation carries the exact build

When a user picks a build from the dropdown, the chosen asset's URL travels with the selection and is treated as authoritative. An earlier version passed only the display label — `“v0.1.2 (zip) (BlackAndWhiteHeadTrack)”` — which matched no release, so the resolver fell through to matching the target game's name and reported “no auto-download found” for a mod it had already located. The game object was also being stripped of its API and bitness on that path.

13. dgVoodoo2 and legacy DirectX

dgVoodoo2, by Dege, wraps DirectX 1–9 and Glide onto Direct3D 11 or 12. It converts an old game into something the modern stereo drivers can work with.

The package

```
MS\x86\       D3D8.dll, D3D9.dll, D3DImm.dll, DDraw.dll
MS\x64\       D3D9.dll only          (DX7/8 predate 64-bit gaming)
MS\arm64x\, MS\arm64_chpe_x86\
Cpl\arm64\dgVoodooCpl.exe            (ARM build of the control panel)
3Dfx\{x86,x64,arm64,arm64_chpe_x86}\  (Glide)
dgVoodooCpl.exe                       (root, x86)
dgVoodoo.conf                         (root)
```

What gets installed

The complete `MS\<bitness>` set is copied beside the game executable, plus the control panel and a DX11-tuned configuration — six files. Not just the single DLL matching the detected API: a game reported as DX9 will often also create a DirectDraw or Direct3D 8 device for intros, movie playback or older subsystems, and dgVoodoo can only wrap what it is present for. The Glide and ARM trees stay out.

```
D3D8.dll   D3D9.dll   D3DImm.dll   DDraw.dll   dgVoodooCpl.exe   dgVoodoo.conf
```

Two bugs found in the real package

- **The control panel could be the ARM64 build.** A plain recursive search for `dgVoodooCpl.exe` finds two copies and directory order decides which, so an x86 machine could silently receive an ARM binary that will not launch. The resolver now prefers the root copy, then an explicitly matching architecture, and refuses anything with `arm` in its path.
- **The generic install path copied everything.** Routing dgVoodoo2 through the normal mod installer dumped 28 files into the game folder, including the entire ARM64 Glide tree. It now redirects to its own installer, which places six.

DX9 is a choice, not a default

The one-click recommender originally treated DX7, DX8 and DX9 identically as “legacy, convert first”. But DX9 has native options — SuperDepth3D, Geo3D, wiz3D and 3DVision4All all run on it directly. Only DX7 and DX8 genuinely have no modern route, so only they default to conversion. The converted pipelines remain available for DX9 as a deliberate choice.

Detected API	Recommended path
DX9	wiz3D or another native driver
DX8	dgVoodoo2 → geo-11 (conversion)
DX7	dgVoodoo2 → geo-11 (conversion)

And there is a genuine conflict on DX9. ReShade proxies `d3d9.dll`, and dgVoodoo2’s wrapper is *also* called `D3D9.dll` — the same filename, a different DLL. Installing dgVoodoo over the top silently replaced ReShade’s proxy and broke both. The installer now checks the slot first and refuses with the correct order of operations: uninstall ReShade, convert, then reinstall it, at which point it uses the DX11 path.

Conversion pipelines

Pipeline	Steps	For
legacy_geo11	dgVoodoo2 → geo-11	Best image on a converted DX7/8 title
legacy_geo3d	dgVoodoo2 → Geo3D	Geometric 3D with ReShade composition
legacy_sd3d	dgVoodoo2 → ReShade → SuperDepth3D	Widest compatibility

The runner marks the game as converted after a successful dgVoodoo2 step so later steps see it as DX11, and stops the pipeline if conversion fails rather than attempting a DX11 mod on an unconverted DX9 game.

14. VR export add-ons and hosts

Export add-ons

Add-on	Pairs with	Sends
SuperVrExport	SuperDepth3D	Full-resolution SBS to KatangaVR or Osiris
GeoVrExport	Geo3D	Full-resolution SBS via the 3DToElse pipeline

Both are ReShade add-ons. They exist because the alternative — letting the headset viewer capture the game window — halves horizontal resolution, since a side-by-side frame in a window of normal width gives each eye half the pixels. The add-on hands over a full-resolution frame instead.

Selecting a VR output format automatically pulls in the matching add-on. This is asserted by the test suite: choosing a VR format without the add-on present produces a black headset view with no obvious cause, which is a particularly unhelpful failure.

Host frameworks

Host	Purpose	Used by
ReShade	Add-on host and shader pipeline	SuperDepth3D, Geo3D, both export add-ons
BepInEx	Unity mod loader	itsloopyo mods on Unity games
ASI Loader	Generic .asi loader	BerZerker mods — usually bundled with the mod itself
REFramework	RE Engine mod framework	Mods for Resident Evil and related titles

Hosts are installed automatically when a mod requires them and are tracked as dependencies, so that removing a driver does not strand its host and removing a host is ordered after the mods inside it. The BerZerker hub mods bundle their own ASI loader, so the application does not add a second one — two ASI loaders in one folder is a reliable way to get nothing loading at all.

15. API compatibility matrix

Mod	DX7	DX8	DX9	DX10	DX11	DX12	Vulkan	OpenGL
SuperDepth3D	●	●	●	●	●	●	●	●
geo-11	·	·	·	·	●	·	·	·
Geo3D	·	·	●	●	●	●	·	·
Legacy Geo3D	·	·	●	●	●	●	·	·
wiz3D (iZ3D wrapper)	●	●	●	●	●	·	·	●
3DVision4All (oneup03)	●	●	●	●	·	·	·	·

● supported · not supported

Every API has a route

API	Direct options	Via dgVoodoo2
DX12	SuperDepth3D, Geo3D	—
DX11	SuperDepth3D, geo-11, Geo3D	—
DX10	SuperDepth3D, Geo3D, wiz3D, 3DVision4All	geo-11, Geo3D
DX9	SuperDepth3D, Geo3D, wiz3D, 3DVision4All	geo-11, Geo3D
DX8	SuperDepth3D, wiz3D, 3DVision4All	geo-11, Geo3D
DX7	SuperDepth3D, wiz3D, 3DVision4All	geo-11, Geo3D
Vulkan	SuperDepth3D	—
OpenGL	SuperDepth3D, wiz3D	—

Vulkan rests on SuperDepth3D alone. That is why it is the safety net and why its API list is deliberately unrestricted. Any change narrowing it would strand Vulkan titles entirely, so this is verified whenever the gates change.

Forceable gates

wiz3D and 3DVision4All install with a warning rather than a refusal when the detected API does not match. Both are proxy wrappers that hook whatever device the game actually creates, so a mis-detected API should not block an install that would have worked. Geometric drivers stay hard-gated: installing geo-11 on a DX12 title cannot work, and pretending otherwise wastes the user’s time.

Where the API comes from

Read from the chosen executable’s import table, cross-checked against the game database where the title is known. The user can override both API and bitness from the game panel, and the override is remembered — necessary for the occasional title whose imports do not reflect what it actually creates at runtime.

16. Output formats

Seven mods offer a choice of stereo output. The selection is written into the mod's real configuration file at install time, so the choice takes effect on first launch.

geo-11 — 7 options

`katanga_vr`, `sbs`, `tab`, `interlaced`, `checkerboard`, `nvidia_dx11`, `nvidia_dx9`

wiz3D (iZ3D wrapper) — 6 options

`interlaced`, `sbs`, `checkerboard`, `anaglyph`, `shutter`, `sr_weave`

3DVision4All (oneup03) — 7 options

`sbs`, `tab`, `row_interlaced`, `column_interlaced`, `checkerboard`, `leiasr`, `katanga`

Geo3D — 7 options

`vr_addon`, `sbs`, `tab`, `interlaced`, `column_interlaced`, `checkerboard`, `anaglyph`

SuperDepth3D — 6 options

`vr_addon`, `sbs`, `tab`, `interlaced`, `column_interlaced`, `checkerboard`

geo-11 (GitHub mirror) — 7 options

`katanga_vr`, `sbs`, `tab`, `interlaced`, `checkerboard`, `nvidia_dx11`, `nvidia_dx9`

Legacy Geo3D — 7 options

`vr_addon`, `sbs`, `tab`, `interlaced`, `column_interlaced`, `checkerboard`, `anaglyph`

What the formats mean

Format	Description
sbs	Side-by-side; the common input for VR viewers and 3D displays
tab	Top-and-bottom; used by some displays and capture pipelines
interlaced / row_interlaced	Alternating scanlines; passive polarised displays
column_interlaced	Alternating columns; some lenticular and autostereo panels
checkerboard	Chequered pixel pattern; older 3D televisions
anaglyph	Colour-filtered; red/cyan glasses, no special display needed
shutter	Active shutter glasses with a compatible display
sr_weave	Simulated-reality weave, for Leia and similar autostereo panels
leiasr	LeiaSR native output
katanga_vr / vr_addon	Full-resolution frame to a VR viewer via the export add-on
nvidia_dx9 / nvidia_dx11	Drives NVIDIA 3D Vision directly

All 47 combinations across the seven mods are exercised by the test suite, which verifies the choice is reflected in the mod's real configuration or preset rather than only in the interface. Roughly a quarter are correctly refused rather than installed — a DX9-only output on a DX11 host, for instance — which is the right answer, not a gap.

17. Configuration handling per mod

The application edits the mod's genuine configuration file in place. It does not keep a parallel copy, because a mod that writes its own settings at runtime would then disagree with the interface, and the user would have no way to tell which was authoritative.

Mod	File	Format	Sections
SuperDepth3D	ReShadePreset.ini	ReShade preset	[SuperDepth3D.fx]
geo-11	d3dxdm.ini + d3dx.ini	INI	[Stereo], [Device], [Rendering], [Loader]
Geo3D	ReShade.ini + preset	INI	[Geo3D], [3DToElse.fx]
wiz3D	wiz3D_Config.xml	XML attributes	GlobalSettings, Keys, Outputs, Presets
3DVision4All	3dvision4all.ini	INI	[stereo], [render], [debug]
dgVoodoo2	dgVoodoo.conf	INI	[General], [DirectX], [GLIDE]
ReShade	ReShade.ini	INI	[GENERAL], [OVERLAY], [INPUT]
BerZerker 6DOF	HeadTracking.ini	INI	[Pose], [FOV], [Network], [Position]
BerZerker 3D+6DOF	"3D-6DOF config.ini"	INI	+ [Stereo3D], [Rotation], [ZAxis], [CullGuard]
itsloopyo	BepInEx .cfg	INI-like	[Connection], [Sensitivity], [Tracking]

Guarantees

- **Edits round-trip.** Writing a value and reading it back returns the value written. This is asserted for every mod, on every engine layout, in several suites — it is the most repeated assertion in the codebase, because a configuration editor that silently fails to save is worse than not having one.
- **Unknown keys survive.** A key the application has never heard of — added by hand, or introduced by a newer version of the mod — is preserved through an edit and through a mod upgrade. Verified by writing a custom key, upgrading the mod, then reading it back.
- **Defaults seed without overwriting.** Installing seeds missing sections so settings are editable before first launch, but never replaces a value the user has already set.
- **Adopted installs are respected.** A mod already on disk that the application did not install can be adopted; its configuration is read, not replaced, and de-registering it does not delete files the application never placed.

Format handling

Three parsers. INI handling preserves comments, ordering and unknown sections. The ReShade preset format is INI-like but uses technique names as section headers, which the shader mods depend on. wiz3D stores values as XML attributes rather than element text, which a generic reader would miss entirely — a test fixture written in the wrong shape produced 46 false failures before being corrected against the real file.

Profiles

A configuration can be saved as a named profile and applied to another game. This is how a user carries a working separation and convergence setup across a series of titles on the same engine.

18. Conflict resolution and proxy slots

A DirectX proxy DLL works by having the name the game loads. Only one file can have that name. This is the largest source of genuine incompatibility between mods, and no amount of clever installation can make two drivers share one entry point.

Contended slots

```
d3d8.dll   d3d9.dll   d3d10.dll   d3d11.dll   d3d12.dll   dxgi.dll
ddraw.dll  opengl32.dll   nvapi.dll / nvapi64.dll
dinput8.dll   winmm.dll   version.dll   dsound.dll
```

The first row is claimed by stereo drivers and wrappers. The second by NVIDIA shims. The third is the generic proxy set that loaders and head-tracking mods use — and because several mods accept any of those four, this is where rehoming becomes possible.

Three resolutions, in order of preference

1. Rehome the incumbent

ReShade can hook through either `d3d11.dll` or `dxgi.dll`; they are equivalent for it on DX10, DX11 and DX12. geo-11 can only be `d3d11.dll`. When geo-11 arrives and ReShade holds that slot, ReShade is renamed to `dxgi.dll` and both end up installed. No reinstall, no download, no user intervention.

A subtle bug in this. The rename updated the manifest, but the install that triggered it had already read its own copy and wrote that back afterwards, silently discarding the change. Uninstalling ReShade then left an orphaned `dxgi.dll` for the next mod to trip over. The rename is now recorded in a sidecar file that no later manifest write can clobber, and uninstall consults it.

2. Move the newcomer

3DVision4All ships four interchangeable proxy names. If its preferred `dinput8.dll` is taken — commonly by a 6DOF ASI loader, which uses the same slot — it installs as `winmm`, `version` or `dsound` instead.

3. Refuse, with the reason and the remedy

Two stereo drivers genuinely cannot share an entry point. The refusal names the incumbent and states the order of operations rather than saying “conflict” and leaving the user to work it out.

```
dgVoodoo2 needs D3D9.dll, but ReShade already owns that file for this game.
On a DirectX 9 title both want the same entry point, so they cannot coexist.
Uninstall ReShade first, convert with dgVoodoo2, then reinstall it - it will
use the DirectX 11 path afterwards.
```

Invariants, asserted everywhere

- One owner per file, across every mod installed into a game.
- One owner per proxy slot.
- Removing any mod leaves every other mod complete on disk and still configurable.
- These hold across 264 mod pairs in both install orders, 72 four-mod stacks in every permutation, 1,512 triple orderings and 219 uninstall permutations.

Why order is tested in both directions

Slot claiming is order-sensitive. A bug that only appears when the head-tracking mod installs first would otherwise hide completely. Every pair is therefore exercised both ways round, and every stack in every permutation.

19. The download layer

Several techniques here were adapted after studying PCVR-Mods-Installer-Hub, whose PowerShell downloader had solved problems this application had not.

Mirrors

Every GitHub asset URL is paired with a Wayback Machine fallback — `https://web.archive.org/web/0/<url>` — tried in order, with any partial file deleted between attempts. It costs nothing when the primary works and rescues a download when GitHub is unreachable or rate-limiting an address. No mirror is invented for non-GitHub hosts, and the mirror is never touched when the primary succeeds; both are asserted.

Archive validation

A CDN answering HTTP 200 with an HTML error page is common, and accepting it produced a confusing failure much later at extraction, reported as “no auto-download found”. Magic bytes are checked before a download is accepted: `PK` for zip, `37 7A BC AF 27 1C` for 7z, `Rar!` for rar. A mismatch falls through to the next mirror rather than being treated as success.

Filename re-resolution

When every mirror fails, the owner, repository and filename are parsed out of the dead URL and the releases feed is searched by name, newest first: exact filename, then the same leading word plus the same extension, then a release whose only asset carries that extension. Anything ambiguous is refused rather than guessed at — three unrelated archives produce no answer, which is asserted. This rescues the case where an author renames or re-tags a release and every pinned URL breaks.

Safety re-copy

After placement, if the rules matched nothing at all, the extracted tree is searched for the folder holding the most payload files — `.dll .asi .addon .fx .fxh .ini .xml .cfg .exe .json` — and the copy is retried from there. This is the net for an author restructuring an archive, which is precisely what produced the nested SuperDepth3D path. An install that writes no files but reports success is a particularly bad failure, because nothing looks wrong.

Timeouts

The original flat sixty-second cap was the worst of both worlds: it killed genuinely large downloads while still letting a dead socket freeze the interface for a full minute. It is now a twenty-second inactivity timeout, re-armed on every chunk received. Large downloads run as long as they need; a stalled connection dies in twenty seconds.

Two crash-class bugs found here. First, redirects were followed with no depth cap, so a host redirecting in a loop recursed until the process ran out of memory — the test run did not fail, it OOM-killed Node with “Ineffective mark-compacts near heap limit”. Capped at five, matching the JSON path which had always had a cap. Second, the inactivity timer stayed armed after a transfer completed, logging a spurious failure exactly twenty seconds after every successful download and calling destroy on a finished socket.

Progress reporting

Progress carries megabyte counters, a live transfer rate and an estimated time remaining, throttled to about four updates per second. The first chunk and a final hundred-percent event are always emitted — without that, a download completing inside one throttle window reported nothing at all.

```
[DOWN] PeakHeadTracking-v1.1.2-nexus   47%   1.5 / 3.2 MB - 68 KB/s - 25s left
[WORK] Extracting PeakHeadTracking-v1.1.2-nexus...
```

This matters more than it sounds. A 3.2 MB mod took 45 seconds at 68 KB/s in a real log, and without numbers on screen that is indistinguishable from a hang. The manual 6DOF install path was passing null for the progress callback, so the one route that actually downloads a per-game mod showed no feedback whatsoever.

Failure modes, all covered by test

Condition	Behaviour
Truncated body	Rejected at validation, next mirror tried
Corrupt archive	Rejected at validation, next mirror tried
Empty body	Rejected; no phantom manifest entry, no stray payload
404 / 500	Next mirror, then filename re-resolution
Rate limited	See section 20
Hung socket	Dropped after twenty seconds of silence
Redirect loop	Capped at five, reported clearly
Disk full	Refused before starting if under 64 MB free
Read-only folder	Preflight write probe fails with an actionable message
Locked file	EBUSY reported as “the game is still running”
Path too long	ENAMETOOLONG reported with the remedy
Antivirus quarantine	Post-write size check detects it and names the likely cause

Environment failures

Node reports these as bare errno strings, which surfaced to the user as “EPERM” and nothing else. Each is now translated into something actionable, a preflight check catches the common ones before anything is written, and any environment failure rolls back the partial install so a retry starts from a clean folder.

Antivirus detection

Antivirus routinely deletes a freshly written `.asi` milliseconds after it lands, and `copyFileSync` has already returned success by then. The install looked perfect and the mod simply never loaded, with nothing to explain why. Every copy now verifies the destination size matches the source; a mismatch names antivirus as the likely cause and suggests adding an exclusion.

20. GitHub without the rate limit

Unauthenticated api.github.com allows sixty requests an hour. A library of a hundred games asks for far more in a single refresh, so the API stops answering and every mod list comes back empty.

The way out

GitHub serves the same information over endpoints that are not part of that quota. They are ordinary CDN-cached web pages, not API calls.

```
/<owner>/<repo>/releases.atom                        every release, as Atom
/<owner>/<repo>/releases                             the HTML release list
/<owner>/<repo>/releases/expanded_assets/<tag>  one tag's assets, as HTML
/<owner>/<repo>/releases/tag/<tag>                   one release, as HTML
```

Four measures

- **Disk cache.** API answers are stored with a ten-minute TTL. Five identical lookups cost one request rather than five — verified.
- **Stop asking a refusing API.** On the first 403 or 429 the reset window is recorded from GitHub’s own `x-ratelimit-reset` header and the API is skipped until it passes. Zero further requests, verified.
- **The block returns a 403-shaped result rather than throwing.** This mattered: an exception skipped every caller’s existing fall-through logic and turned a recoverable throttle into a dead end. Returning the shape the callers already handle keeps the HTML fallbacks working unchanged.
- **Stale cache beats nothing.** If the API is blocked and no fresh entry exists, an older cached copy is returned rather than an empty list.

The fallback must run when limited, not instead of

One guard read `if (!versions.length && !limited)` — so the HTML scrape was skipped in exactly the case it exists for. The symptom was one source returning versions while the other returned zero in the same session, which is what pointed at it.

Honest messaging

“Rate-limited — try later” appeared even when the fallback had succeeded. The throttle is now reported only when the application genuinely came up empty. Settings shows the live state, offers a manual cache refresh, and notes that a personal access token needs no scopes ticked at all.

```
[OK]   API available - results cached for 10 minutes
[WARN] API throttled for another 12 min - using the public release pages
       and cache instead, so mods still list.
```

A token raises the limit from sixty to five thousand requests an hour and avoids the situation entirely, but the application is designed to work without one.

21. Update machinery and self-update

Per-mod updates

Each core source is checked against its own discovery strategy — GitHub releases, a blog feed, a website scrape or a pinned version. A cached core carries the version it was fetched at, and updating it re-links every game using it rather than re-downloading per game.

A branch-tracking subtlety. SuperDepth3D is fetched from a branch, so its cache is named `master` while the update check reports `master@<sha>`. A plain string comparison can never match, so the update badge was stuck on permanently no matter how many times the user updated. Comparison is now by commit, and a branch cache with no recorded commit reports not outdated — nagging with an update the user cannot clear is worse than occasionally missing one.

Application self-update

Electron cannot overwrite its own running files on Windows, so the swap is handed to a batch script. The application downloads the new build, unpacks it to a staging folder beside itself, writes the script, launches it detached and quits.

```
wait for the app's PID to exit         60s cap, then abort changing nothing
+3s grace so Windows releases handles
robocopy /E /IS /IT /R:3 /W:2
  /XF update.bat  /XD logs\  manual-core\  .update-staging\
if errorlevel 8 -> log, relaunch the OLD build, change nothing
otherwise       -> relaunch, remove staging, delete the script
```

Everything is appended to `logs\update.log`. An installer-style `.exe` asset is refused with an explanation rather than unpacked over the application, and a missing download is refused rather than producing a half-written script.

The standalone updater

`update.bat` ships alongside the executable for updating with the application closed. It refuses to run while the application is open, asks GitHub for the newest release, downloads and unpacks it, unwraps the release folder, aborts if the archive contains no application executable, and excludes itself, the logs and manual-core from the copy.

Build fingerprint

Every build carries an eight-character hash of the source files, logged at startup and shown in Settings. This exists because several debugging rounds were spent on symptoms that turned out to be a stale build — identical log timestamps across three separate reports eventually revealed the application had not been restarted. “Which build is running” is now answerable from one log line.

```
[INFO] versions {"app":"1.0.0","build":"86fc41b8","built":"...", ...}
```

`stamp-build.js` regenerates it and is wired into the packaging script, so a release cannot ship a stale stamp.

22. Logging

Two files in a logs folder beside the executable — the same place manual-core lives, so both are findable without navigating to AppData.

File	Contents
download.log	Every request: URL, redirects, status, bytes, duration, rate, and the exact failure reason
app.log	Startup and versions, every IPC call, core resolution, extraction, placement, config writes, manifest changes, uninstalls, crashes
update.log	Written by the update scripts only

Both rotate at 5 MB keeping one previous generation, so they can be left on permanently. Logging never throws — every write is wrapped, because a logging failure must not break an install.

What a successful install looks like

```
INFO install sd3d        {"game":"Red Dead Redemption","api":["DX11"],"bit":"x64"}
INFO core resolve sd3d
INFO core resolve reshade
INFO placement           {"mod":"SuperDepth3D","from":"Shaders","files":1}
INFO manifest write      {"mods":["reshade","supervrexport","sd3d"],"files":6}
INFO install OK sd3d     {"ms":13,"tag":"manual"}
INFO config write sd3d   {"sections":["SuperDepth3D.fx"]}
```

The placement line is the one that would have shown the nested SuperDepth3D path immediately, rather than after a screenshot of a file explorer.

Outcomes are logged, not just attempts

The logger originally recorded only that an install had started. A failure was indistinguishable from a success, which made a user-supplied log nearly useless. Every install now logs its result with the reason.

```
ERROR install FAILED geo11  {"note":"geo-11 does not support Vulkan (it drives DX11)"}
```

Download detail

```
START 6DOF Hub (BerZerker96) - Mod-v1-nexus.zip   {"url":"..."}
REDIR redirect 1                                  {"from":"...","to":"..."}
HTTP  HTTP 200                                    {"contentLength":267}
OK    ... complete                                {"bytes":267,"ms":4,"kbps":65}
FAIL  ... FAILED                                  {"error":"HTTP 500 for ..."}
```

Secrets

Any key matching `token`, `password`, `secret` or `authorization` is written as `[redacted]`. Asserted by test: writing a value of `ghp_secret` must not appear on disk. This is what makes it safe to ask a user to attach a log to a bug report.

In-application access

Settings offers Open logs, a viewer for the last sixty lines of either file, and Clear.

23. Persistence and settings

File	Contents
settings.json	Theme, scan roots, exclusions, GitHub token, icon scale, window bounds, background pattern, ambient glow, detail width
library.json	Every detected game: name, folder, executable, executable directory, API, bitness, engine, installed and detected mods
profiles.json	Saved per-game mod configurations
<game>\.stereoscope\manifest.json	Exactly what this application placed in that game, per mod, with tags and timestamps

The manifest is the contract

Every install records the exact list of files it placed. Uninstall removes that list and nothing else. This is what makes it safe to install into a game folder at all — the application never guesses at what belongs to it, and never deletes a file it did not write.

File lists are deduplicated centrally on write. geo-11 ships parallel 32- and 64-bit trees that collapse onto the same destination, and recording a file twice corrupted ownership accounting and made uninstall attempt to delete the same path twice. Deduplicating in one place, rather than at each of several write sites, is what made the fix hold — an earlier attempt patched only one of them.

Migration

Settings, profiles and library are migrated once from the old location beside the executable. Originals are renamed with a `.migrated` suffix rather than deleted, so nothing is lost if the migration misbehaves.

Library pruning on load

Nested entries left by older builds are removed when the library loads, and the cleaned result is saved. This is necessary because a scan merges rather than rebuilds, so a scanner fix does not repair rows that were already saved.

24. The user interface

Library

A grid of detected games with an ambient glow behind each icon sampled from its artwork. Each card shows the rendering API, bitness and badges for detected mods. A red gear on the icon opens per-game mod configuration.

Filtering

- A search box filtering by name as you type, with a live match count.
- Drive rows generated from the loaded library with real per-drive counts, clickable to filter.
- Render API, bitness and “has a mod for it” chips.

The drive rows were previously four hardcoded placeholders with invented counts and no click handler, which is why selecting a drive did nothing — there was nothing there to select.

Scanning

One button with a dropdown. Every entry — including “All drives” — calls the same function with one argument, so a per-drive scan cannot diverge from the all-drives scan that is known to work. Each drive row shows how many games the library already has there. The menu is a body-level fixed panel rather than nested in the toolbar, because nested it was painted over by later siblings and did not read as a dropdown at all.

All-exes mode

A toggle beside the scan control. When on, a scan adds every executable found in each game folder as its own library entry rather than only the best candidate, each with its own API and bitness read from that binary. Shift-clicking opens a one-off picker for the selected game, listing every executable with real binaries first and tools marked, so detection can be overridden directly. This is the escape hatch for cases detection cannot reasonably solve.

Game detail

A one-line summary carries the two things that matter — rendering API and the recommended mod — with the full scanner result folded behind an Advanced info toggle whose state persists across selections. Play, Open folder, Rescan mods and Remove all mods sit above it.

Play launches the game detached from its own folder so data paths resolve correctly. Open folder opens the directory the executable is in, not the game root — for `Judgment\runtime\media` that is where every mod file goes.

Remove all mods

Lists what will be removed before doing it, and removes in dependency order — guests before hosts, since ReShade hosts SuperDepth3D and the export add-ons, and removing it first would strand them. Each mod is removed independently so one failure cannot abort the rest.

One-click setup

A pipeline dropdown per game offering the recommended route and its alternatives, with the output format as a second dropdown. The recipe is shown in full before installation so the user knows exactly what is about to happen.

Mods tab

An index of every project the application builds on, organised into six sections with this application pinned and highlighted at the top, followed by the managed mod list with cache status, sizes and per-mod actions.

Guide

Five side-by-side pages: Start Here, depth-based 3D, geometric 3D, wrappers, and 6DOF. Includes an API comparison table, accurate per-mod hotkeys taken from shipped configuration files rather than documentation, and the decision rule for choosing a driver.

25. The test suite

Twenty-one suites, over seventy-one thousand assertions, all run against a deterministic HTTPS mock and a sandboxed filesystem. No test touches the network or a real game folder.

Suite	Checks	Covers
smoke	36	Syntax, IPC pairing, registry integrity, bundled payloads, scanner layouts
matrix	258	Every mod against every engine layout
exhaustive	11,259	Combination simulation across the catalogue
updates	710	Version discovery, self-update, UI contract, link index, logging
configs	322	Config round-trips, hotkeys, preinstalled sections
sixdof	430	6DOF installs and API reversion
total	41,009	1,210 pairs, 1,512 triple orderings, 219 uninstall permutations
manual6dof	4,180	Every game, every version, both sources
quickpath	643	One-click pipelines across every layout
permod	121	Each mod audited individually, end to end
everymod	6,552	All 80 6DOF mods, every version, plus outputs, stacks and network faults
fullmatrix	4,053	3D x 6DOF x API, with and without dgVoodoo2
exematrix	365	103 real executable folder shapes, plus the demote lists both ways
pcvraudit	39	Mirrors, safety re-copy, filename re-resolution, progress
envfail	20	Permissions, locked files, disk full, antivirus quarantine
dgvupdate	66	dgVoodoo2 caching and the self-update script

What the large suites assert

- **Every install reaches a decision.** Never a crash, never a silent no-op. A refusal is a valid outcome; an exception is not.
- **Refusals and acceptances are both justified.** A mod refused on API grounds must genuinely not support that API, and a mod accepted must genuinely support it.
- **State does not drift.** Install, uninstall and reinstall three times: the file footprint and configuration shape must be identical on each pass, with no duplicate records and nothing left behind between passes.
- **Order does not matter.** Pairs in both orders, four-mod stacks in every permutation, uninstalls in every order, with survivors verified complete after each removal.
- **Adversity is handled.** Eight network fault types against both 6DOF sources, each asserting no phantom manifest entry and no stray payload — a half-written install that looks installed is worse than a failed one.

Fixture accuracy

Several rounds found bugs in the fixtures rather than the application: a wiz3D configuration written in the wrong XML shape, a zip labelled as a 7z, a helper that bypassed a rule the application correctly enforces. These are recorded because a suite that drifts from reality gives false confidence. Fixtures are now built from real artefacts — an actual dgVoodoo2 package, an actual combined 3D+6DOF build, an actual 117-game library.

Running them

```
bash harness/validate.sh          # all 21 suites in order
node harness/exematrix.js         # executable detection only
node harness/updates.js           # update machinery and UI contract
node harness/fullmatrix.js        # 3D x 6DOF x API combinations
```

26. Bug ledger: what was found and fixed

Recorded because the same mistakes are easy to reintroduce, and because the pattern in them is instructive.

Area	Bug	Root cause
Downloads	Redirect loop exhausted memory	No depth cap; the JSON path had one, download did not
Downloads	Spurious failure 20s after every success	Inactivity timer stayed armed after completion
Downloads	Corrupt HTTP 200 accepted as success	No magic-byte validation before accepting
Downloads	Manual 6DOF showed no progress	Null passed for the progress callback on that path
Placement	SuperDepth3D nested three levels deep	Wrapper folder not unwrapped; copied the whole repo
Placement	dgVoodoo2 dumped 28 files inc. ARM64 Glide	Routed through the generic installer
Placement	dgVoodoo2 control panel could be ARM64	Recursive search; directory order decided
Placement	geo-11 recorded duplicate manifest entries	Parallel 32/64 trees collapsing onto one path
Scanner	Tools chosen over games, repeatedly	Name blocklist only; no binary or size signal
Scanner	Subfolders registered as separate games	No nested-candidate deduplication
Scanner	A game named “Windows”	Folder name taken literally
Scanner	Root executable chosen over the real one	Candidate list returned on first match
Scanner	Per-drive scan found nothing on Steam drives	Skipped the explicit Steam library enumeration
Scanner	Win64MasterMasterSteamPGO not recognised	Binary folder matched exactly, not by prefix
Library	A failed scan wiped the library	Array cleared before the result was examined
Library	Stale bad rows survived every fix	A scan merges; it never re-derives existing rows
Config	Combined 3D+6DOF stereo settings invisible	Section allow-list missing Stereo3D and four others
Config	dgVoodoo2 had no editor	Declared defaults but no section list or path resolution
Conflicts	Orphaned dxgi.dll after rehoming	Manifest write raced the in-flight install
Conflicts	dgVoodoo2 silently replaced ReShade on DX9	No slot check before writing
UI	Settings tab showed the previous page	Undefined global threw inside the render function
UI	Scan dropdown painted under the game cards	Nested in the toolbar; stacking context
UI	Drive rows did nothing	Hardcoded placeholder markup with invented counts
UI	Search placeholder showed a literal escape	Escape sequence in a plain HTML attribute
UI	6DOF picker stuck on “loading…”	Apostrophe in a game name broke a quoted attribute
Network	itsloopyo empty while BerZerker worked	Scrape fallback skipped when rate-limited

Area	Bug	Root cause
Network	“Rate limited” shown after success	Flag reported regardless of the outcome
Network	Pre-release mods invisible	/releases/latest excludes pre-releases and 404s
Network	Pre-release assets excluded	Asset filter rejected the only asset present
Updates	Update badge permanently stuck	Branch cache compared against branch@sha
Icons	No artwork off the Steam drive	Cache path derived from the game, not from Steam
Recommend	dgVoodoo2 forced onto every DX9 game	DX7/8/9 treated identically as “legacy”

The pattern

Three successive rounds of widening the synthetic combination matrix found one bug, then one, then none. Meanwhile a single screenshot, a single log file and a single real library file each found several. Real artefacts are worth more than more combinations. The fixtures are now built from them, and the highest-value input for future work is a real log or a real folder listing rather than another permutation.

Recurring mistake classes

- **Blanket string replacement in a large file.** Two bugs came from replacing a pattern without checking the surrounding structure — one turned a guarded return into an unguarded one in eight places at once.
- **Returning early from a candidate list.** The scanner did this twice, in different functions, with the same result: a plausible-but-wrong answer that stopped the search.
- **Fixing derivation without fixing saved state.** A scanner fix does not repair a library that already contains bad rows.
- **Reporting a warning regardless of outcome.** The rate-limit flag and the update badge both did this; both read as failures when the operation had succeeded.

27. Known gaps and unverified areas

Everything below is either untested, environment-limited, or a deliberate trade-off. This section exists so whoever inherits the work does not have to rediscover it.

Verified by inspection, not execution

Item	Why	Risk
The update .bat swap	Development is on Linux; tasklist and robocopy do not exist there	Medium — written defensively, aborts rather than half-copying, relaunches the old build on failure
Read-only folder handling	The container runs as root, which ignores chmod	Low — verified by errno injection instead
RAR extraction	No rar packer available to build a fixture	Low — zip and 7z verified end to end
Elevated helper for 3DVision4All	Requires a real UAC prompt	Low

Known behavioural limits

- **Emulators in game folders.** A folder called “Persona 5 Royal” containing `Ryujinx.exe` resolves to the emulator. Nothing in the binary or the path says otherwise. Use the exclusion list or the All-exes picker.
- **Name-based artwork lookup.** Folder names like `VaM_1.20.77.9` or `CCFF7R` will not resolve to a store page.
- **Rendering tools.** Creation Kit and texconv genuinely link D3D11, so graphics scoring alone cannot demote them. Size banding handles them, with the name list as a backstop.
- **Scan duration.** Whole-tree ranking walks to depth five per game folder. The graphics test is cached per file and reads only the first six megabytes, but a first full scan of a large library is slower than a shallow one.
- **wiz3D TAB and column-interlaced.** Pending upstream confirmation of the correct output DLL names.
- **Broken Sword Remastered** reports DX9 where DX11 is expected. The executable is correct, so this is a detection nuance rather than a wrong pick; the API dropdown overrides it.

Deliberate trade-offs

- **A branch core with no recorded commit reports not-outdated.** Occasionally missing an update is better than a badge the user cannot clear.
- **The rate-limit block is global, not per-repository.** GitHub’s quota is per address, so one 403 means every repository is throttled and there is no value in asking again.
- **Tool names are a penalty, not an exclusion.** A folder of oddly-named binaries still yields an answer rather than nothing.
- **ReShade backup files survive Remove all mods.** Deleting them would destroy the ability to restore the pre-mod state.
- **Multiplayer titles are not blocked, only warned about.** The decision belongs to the user, but the risk is stated plainly at the point of installation.

If something is wrong, what to send

A folder listing of the affected game, the relevant section of `app.log`, and the build fingerprint from Settings. Those three between them have resolved every issue faster than any amount of reasoning from a description.

28. Operational runbook

Building

```
npm install
node stamp-build.js      # refresh the build fingerprint
npm run dist             # package (runs the stamp automatically)
```

Diagnosing a user report

Symptom	First look	Then
Wrong executable chosen	library.json: exe and exeDir	Add to the demote list, or use the All-exes picker
Mod will not download	download.log	Check for FAIL lines and whether the mirror chain ran
Install did nothing	app.log: install OK / FAILED	The note field carries the reason
Mod list empty	app.log: rateLimited	Settings shows the throttle state; add a token
A feature seems missing	Settings: build fingerprint	Compare against the expected build
Game will not launch modded	manifest.json in the game folder	Check for proxy slot contention
Icons missing	app.log around icon resolution	Try Fetch missing icons in Settings
Settings tab blank	DevTools console	An undefined global in a render function kills the page

Adding a mod to the catalogue

- Add an entry to `src/mods.js`: name, source strategy, placement rules, supported APIs, configuration file.
- Add a core source if it needs a shared download, with its discovery strategy.
- Add its configuration sections to `MOD_SECTIONS` in the installer, and a path resolution case if the file is not beside the executable.
- Add a fixture to `harness/permod.js` — the per-mod audit then covers download, placement, configuration round-trip and uninstall automatically.
- Run `validate.sh`. The matrix suites will exercise the new mod against every layout and every other mod without further work.

Adding a game shape to the scanner

- Add the folder pattern to the candidate list in `src/scanner.js` if it is a named, stable shape.
- Add a case to `harness/exematrix.js` with the real folder contents, including whatever decoys ship alongside.
- If the correct executable is losing to a tool, prefer adding the tool to the demote list over adding the game to an allow list — and assert a set of real game names still survive.

Where to be careful

- **Widening a blocklist.** Always assert in both directions; the false-positive half matters more than the coverage half.
- **Blanket string replacement in the installer.** Check the surrounding structure, not just the matched text.
- **Adding UI that references state.** An undefined global inside a render function kills the whole page silently and presents as a routing bug. The suite now asserts none exist.

- **Changing MOD_API.** Verify every API still has at least one working route afterwards; Vulkan has exactly one.
- **Changing the manifest format.** It is the only record of what may safely be deleted from a user's game folder.

Release checklist

- Run the full suite; all twenty-one must pass.
- Run `stamp-build.js` and confirm the fingerprint changed.
- Verify the packaged zip contains `update.bat` beside the executable, not inside the asar.
- Launch, open Settings, confirm the build fingerprint matches.
- Scan one drive and confirm the library merges rather than replaces.
- Install and uninstall one mod, and confirm the game folder returns to its prior state.

End of document. Build 86fc41b8. Generated 2026-08-09.

Appendix A. The complete IPC surface

Every channel the renderer may invoke. 99 handlers, each bridged exactly once in preload. The symmetry is asserted by the smoke suite.

Scanning and the library

`scan`, `scanDrive`, `drives`, `gameInfo`, `listExes`, `getLibrary`, `setLibrary`, `addManualGame`, `removeManualGame`, `getManualGames`, `gameModStatus`

Installing and removing

`install`, `uninstall`, `setupDgVoodoo`, `adoptMods`, `htInstallManual`, `applyDx9Proxy`

Mod discovery

`htVersionsFor`, `htCatalog`, `coreList`, `coreUpdate`, `coreRoot`

Configuration

`readModConfig`, `writeModConfig`, `readModAnalyzed`, `listProfiles`, `saveProfile`, `loadProfile`, `deleteProfile`

Updates

`appUpdateCheck`, `appUpdateDownload`, `appUpdateApply`, `checkUpdates`, `buildInfo`, `appVersion`

Settings

`getSettings`, `setSettings`, `addScanRoot`, `removeScanRoot`

Diagnostics

`logPaths`, `openLogs`, `clearLogs`, `tailLog`, `rateLimitState`, `clearGhCache`

Shell integration

`openFolder`, `openExternal`, `launchGame`, `icon`, `fetchIcon`

Other

`analyzeConfigs`, `checkHeadTracking`, `configPath`, `coreFetch`, `coreFetchAll`, `coreSources`, `detectGame`, `downloadHeadTracking`, `excludeGame`, `exportAllSettings`, `htMatch`, `hubAllReleases`, `hubDownloadAll`, `hubInstallInto`, `hubPooled`, `hubSuggest`, `hubVersions`, `importAllSettings`, `latest`, `list6dofMods`, `manualCoreStatus`, `mods`, `openCoreFolder`, `openDataFolder`, `openHelixFix`, `openHtConfig`, `openManualCore`, `pickCoreDir`, `pickExe`, `postInstallInfo`, `proxyRename`, `proxyState`, `readConfig`, `readConfigFile`, `readHtConfig`, `readModFiles`, `redetectGame`, `resetCoreDir`, `runPostInstall`, `unexcludeGame`, `wizGetConfig`, `wizOutputs`, `wizSetConfig`, `wizSetOutput`, `writeAnalyzed`, `writeConfig`, `writeConfigFile`, `writeHtConfig`, `writeModFiles`

Any handler added without a matching preload binding, or any binding pointing at a handler that does not exist, fails the smoke suite rather than surfacing as a silent no-op at runtime. This has caught real omissions during development.

Appendix B. The game database

3,335 titles with engine, rendering API and bitness hints. Used to cross-check what the scanner reads from a binary, and to recommend a driver before anything is installed.

By rendering API

API	Titles	Share
DX11	1,866	56.0%
DX9	692	20.7%
DX12	450	13.5%
OpenGL	140	4.2%
DX8	70	2.1%
Vulkan	54	1.6%
DX7	50	1.5%
DX10	13	0.4%

The distribution is the reason SuperDepth3D matters so much: DirectX 11 and 12 dominate, and geo-11 covers only one of those. It is also why the legacy conversion path exists at all — the DX7 through DX9 tail is small in percentage terms but represents hundreds of titles.

By engine

Engine	Titles	Engine	Titles
Unity	780	RE Engine	18
Custom	614	TW Engine	18
Unreal Engine 4	480	Katana Engine	18
Unreal Engine 5	182	MonoGame	15
Unreal Engine 3	144	Clausewitz	14
Source	38	Frostbite 3	12
Emulator	37	Dragon Engine	12
Frostbite	29	Apex (Avalanche)	12
CryEngine	28	AnvilNext	12
EGO Engine	27	PhyreEngine	12
MT Framework	24	GoldSrc	11
GameMaker	23	Custom (Java)	11
Unreal Engine 2	19	id Tech 3	11

Engine matters because it predicts the folder layout. An Unreal title puts its binary in `<Project>\Binaries\Win64`; a Unity title puts it at the root beside a `_Data` folder; RE Engine games are flat. It also predicts which mod host is needed — BepInEx for Unity, REFramework for RE Engine.

What a database entry provides

- A canonical display name, so a folder called `P5R` shows as Persona 5 Royal.
- The expected rendering API, cross-checked against the executable's imports.
- The expected bitness.
- The engine, which informs both the scanner and the choice of mod host.

The database is a hint, never an override. What the binary actually imports wins, because a remaster or a patch can change the API without changing the name. Where they disagree the interface shows the detected value with the database value available as a one-click alternative.

Appendix C. Engine layouts recognised

Eleven layouts are modelled in the test harness, each with its real folder shape, rendering API and bitness. Every mod is exercised against every layout.

Layout	API	Executable location	Mod host
Unreal 5	DX12	<Project>\Binaries\Win64*-Win64-Shipping.exe	ReShade
Unreal 4	DX11	<Project>\Binaries\Win64*-Win64-Shipping.exe	ReShade
Unreal 3	DX9	Binaries\Win32 or Win64	ReShade
Unreal 1/2	DX7/8	System\	—
Unity	DX11	Root, beside <Name>_Data\	BepInEx
RedEngine	DX12	bin\x64\ or bin\x64_dx12\	ReShade
CryEngine	DX11	Bin64\ or bin\win_x64\	ReShade
Source	DX9	Root, or game\bin\win64\	ReShade
RE Engine	DX12	Root	REFramework
RAGE	DX12	Root	ReShade
Dragon Engine	DX11	runtime\media\	ASI loader
Vulkan titles	Vulkan	Varies	ReShade (SuperDepth3D only)

The layouts exist so that a placement rule can be verified against the shape a real game actually has. A mod that installs correctly at a game root but not inside `Binaries\Win64` is broken for most modern titles, and only a layout-aware test finds that.

Appendix D. Executable classification, in full

The two lists that decide whether a binary can be the game. Both are asserted in the test suite in both directions — tools rejected, and 115 real game executables kept.

Never picked

Skipped outright. Nothing in this list can ever be a game.

```
(^|[^a-z])(unins\w*|setup|\w*installer|installer\w*|quicksfv|miniconda|vc_?redist|dxsetup|directx|oa
linst|dotnet\w*|prereq\w*|ueprereq|crashpad|crashreport\w*|crashhandler\w*|easyanticheat\w*|battleye
|be_?service|touchup|activation|cleanup|notification|redist)([^a-z]|$)|crashhandler|crashreport|_red
ist
```

Demoted

Sorted last but still eligible, so a folder containing only oddly-named binaries still yields an answer.

```
^(play|start|launch|boot|run)[A-Z0-9]|launcher|benchmark|_loader|server\.exe$|editor\.exe$|devkit\.e
xe$|tool\.exe$|config\.exe$|settings\.exe$|^(skse|f4se|obse|nvse)\d*|dedicated|modmanager|ansel|fbxt
ogranny|texconv|creationkit|bssndrpt|driverversionchecker|^patch\b|windowsdesktop-runtime|dotnet-run
time|^vc_|oalinst|^activation|crashsender|hkxcmd|chardetect|trainer|cheat\s*engine|\bfling\b|nifskop
e|bsa\s*browser|xedit|tes5edit|sseedit|fo4edit|loot\.exe$|wrye|mo2|vortex|resaver|papyrus|bethini|en
binjector|d3dcompiler|dxwebsetup|unrealcefsubprocess|epicwebhelper|steamerrorreporter|gameoverlayui|
dumper|symbols|pdb|profiler|shadercompiler|shaderc|installscript|bugsplat|crash_?handler|saveremover
|pygrun|layerschecker|preprocessor|vrcompositor|wallpaperinject|applicationwallpaper|_tool\d*\.exe$|
tool\d*\.exe$|remover|packer|unpacker|extractor|repacker|decompil|disasm|hasher|validator|verifier|d
iagnos|telemetry|overlay|companion|steamvr|vrmonitor|vrserver|vrdashboard|vrwebhelper|helper\.exe$|h
ost\.exe$|service\.exe$|monitor\.exe$|agent\.exe$|daemon\.exe$|crs-?video|bink|movieplayer|videoplay
er|smackw|umediaplayer|wmvplayer|^criware|^cri_|criware|adx2|scaleform|autorun|register\.exe$|activa
te\.exe$|3dmigoto|migoto\s*loader|nvdxt|nvcompress|cefsharp|cefclient|libcef|leaveingamedir|^ds3t$|d
s3t\.exe$|knobcontroller|companion\s*app|modloader|mod\s*loader|asiloader|scripthook|reframework|^di
nput8\.exe$|steamvr|vrserver|vrmonitor|oculus\w*setup|openvr|shadercompileworker|unrealpak|unreallig
htmass|swarmagent|unrealcefsubprocess|ue\d?prereqsetup|crashreportclient|epicwebhelper|eosbootstrapp
er|epicgameslauncher|easyanticheat|eac_\w*|battleye|be_?service|beclient|punkbuster|pbsvc|pbcl|gameg
uard|npkcrypt|nprotect|xigncode|xhunter|denuvo|vmprotect|themida|vac\.exe$|anticheat|ac_?client|upla
y\w*|ubisoft\w*connect|galaxyclient|gog\w*galaxy|origin\w*setup|eadesktop|rockstar\w*launcher|social
club|bethesda\w*launcher|battle\.net|updater|patcher|repair|verif\w*|downloader|bootstrapper|physx\w
*|xnafx|dotnetfx|dxwebsetup|directx\w*redist|openal\w*|oalinst|wwise\w*|fmod\w*setup|havok\w*|speedt
ree|coherent\w*|awesomium|werfault|breakpad|crashpad_handler|sentry|bugreport|errorreport|feedback|r
tss|afterburner|shadowplay|overwolf|discord\w*hook|fraps|bandicam|^install$|^install\.exe$|^readme|^
manual\.exe$|^credits\.exe$|acroread|adobe\w*reader|^dxdiag|^regsvr|^msiexec|^rundll|^cmd\.exe$|^con
host|^wscript|^cscript
```

Folders never walked

```
^(\$|windows$|windows\.old|system volume|recovery|perflogs|programdata|appdata|boot|msocache|intel$|
amd$|nvidia$|drivers$|node_modules|\.git|temp$|tmp$|cache$|users$|public$|default$|all users|_?redis
t|_?commonredist|redistributable|installer_files|installers?$|vcredist|directx$|dotnet|_mei\d*|__ins
taller|prerequisites?|prereq|dxsetup|support$|extras?$|docs?$|manual$|soundtrack|artbook|savegames?$
|screenshots?$|crashes$|logs?$|dumps?$|tools?$|sdk$|modtools?$|editor$|devkit$|_backup|backup$)
```

Maintaining these. Prefer adding a tool to the demote list over adding a game to an allow list, and always re-run the false-positive assertions. One pattern, `^cri`, was caught that way — it matched **Cri**msonDesert as well as CriWare.

Appendix E. Executable shapes verified

103 real folder structures, each asserted to resolve to the correct binary. Collected from shipping games across two decades.

Unreal Engine

Shape	Example
<Project>\Binaries\Win64	Palworld, Remnant II, Ready or Not, Satisfactory
Project name differs from the game	Deliver Us The Moon → MoonMan\Binaries\Win64
Blueprint-only	Engine\Binaries\Win64\UE4Game-Win64-Shipping.exe
Game Pass / GDK	Content\Game\Binaries\WinGDK
Deeply nested	Game\Content\Binaries\Win64
UE3 flat	Binaries\Win64 or Binaries\Win32
UE1/2	System\UT2004.exe, System\DeusEx.exe

Other engines and publishers

Shape	Example
runtime\media	Judgment, Lost Judgment, Yakuza: Like a Dragon
Game\	Elden Ring, Dark Souls III, Armored Core VI
game\	Final Fantasy XIV → game\ffxiv_dx11.exe
bin\x64	Cyberpunk 2077, The Witcher 3, Factorio
bin\x64_dx12	Cyberpunk 2077 DX12 variant
Bin64\	Crysis 3, Star Citizen
bin\win_x64	Hunt: Showdown
Base\Binaries\Win64Steam	Civilization VI
bin\Win64Master…	Kingdom Come: Deliverance II
pso2_bin\	Phantasy Star Online 2
_retail_\	World of Warcraft, Overwatch
<Name> Game\	Genshin Impact
DefEd\bin	Divinity: Original Sin 2
ph\work\bin\x64	Dying Light 2
Game\ME1\Binaries\Win64	Mass Effect Legendary Edition
game\bin\win64	Dota 2

Flat-root titles

Doom Eternal, Skyrim, GTA V, Red Dead Redemption 2, Resident Evil 2/4/8, Devil May Cry 5, Monster Hunter World, Dragon’s Dogma 2, Horizon Zero Dawn, Control, Alan Wake 2, Metro Exodus, God of War, Escape from

Tarkov, Apex Legends, Destiny 2, Valheim, Subnautica, Terraria, Stellaris, X4, Path of Exile, Arma 3, and many more. These are the easy case, and the reason a root executable must never be dismissed simply for being at the root.

Tie-breaks verified

Case	Winner
64-bit and 32-bit shipping builds present	64-bit
Project build and Engine\ fallback both present	Project build
A shallow bin\ and a deeper one	Shallower
Five launchers and one small real binary	The binary
DX12 and DX11 builds of the same game	DX12
PathOfExile.exe and PathOfExile_x64.exe	The 64-bit build
No executable at all	Nothing, cleanly

Appendix F. Quick reference

Choosing a driver

If the game is…	Use	Because
DirectX 11	geo-11	True geometric stereo, the best image available
DirectX 12	Geo3D, or SuperDepth3D	geo-11 does not cover DX12
Vulkan	SuperDepth3D	The only option
DirectX 9 or 10	wiz3D, 3DVision4All, Geo3D or SuperDepth3D	All work natively; no conversion needed
DirectX 7 or 8	dgVoodoo2, then geo-11	Nothing modern speaks those APIs directly
On an unusual display	wiz3D	Widest output format set: anaglyph, shutter, interlaced, weave
Going to a headset	Any driver plus its VR export add-on	Full resolution rather than a captured window

Hotkeys

Mod	Toggle	Separation	Convergence	Other
SuperDepth3D	via ReShade overlay	In-shader	In-shader	Space opens the overlay
geo-11	—	Per d3dxdm.ini	Per d3dxdm.ini	Community fixes add their own
wiz3D	Numpad *	Numpad + / –	Shift+Numpad + / –	Numpad 7/8/9 presets, Ctrl+F8 swap eyes
3DVision4All	Per 3dvision4all.ini	Per ini	Per ini	—
BerZerker 3D+6DOF	Per config	Per config	Per config	28 keys in [Stereo3D]

Log lines worth searching for

```
install OK          a successful install, with timing and tag
install FAILED      a refusal, with the reason in the note field
install THREW       an exception that crossed the install boundary
placement           what was written, from where, and how many files
FAIL                a download failure, with the HTTP status or error
rateLimited         the GitHub throttle state
core resolve        which source was consulted for a mod
manifest write      what the game folder now records
```

File locations at a glance

```
%APPDATA%\Stereo3D Manager\settings.json     preferences
%APPDATA%\Stereo3D Manager\library.json      detected games
%APPDATA%\Stereo3D Manager\core\            cached mod payloads
%APPDATA%\Stereo3D Manager\cache\github\    release metadata
<app>\logs\app.log                          everything the app did
<app>\logs\download.log                     every network request
<app>\manual-core\<mod>\                    hand-supplied packages
<game>\.stereoscope\manifest.json           what was installed here
```

End of appendices.

Appendix G. Per-mod reference

One page per remaining catalogue entry: what it is, where it comes from, what it places, and anything specific that has caused trouble. The five stereo drivers are covered in section 11.

ReShade (add-on build)

Property	Value
Role	Host framework and shader pipeline
Rendering APIs	DX7, DX8, DX9, DX10, DX11, DX12, Vulkan, OpenGL
Source	reshade.me/
Config file	ReShade.ini
Cached core	yes

Places

```
dxgi.dll  or  d3d11.dll  or  d3d9.dll   (whichever the game needs)
ReShade.ini            global settings
ReShadePreset.ini      the active preset
reshade-shaders\Shaders\   the shader library
```

The add-on host that SuperDepth3D, Geo3D and both VR export add-ons run inside. Installed automatically whenever one of them is chosen, and tracked as their dependency so removal is ordered correctly.

The add-on build is required rather than the plain build — Geo3D and the export add-ons are add-ons, not shaders, and the plain build will not load them.

The overlay key is set to Space at install time rather than the default Home, because Home is bound in a great many games and Space in far fewer while the overlay has focus.

ReShade is the mod most often involved in a proxy conflict, because it can legitimately occupy several different slots. That flexibility is also what makes rehoming possible — see section 18.

geo-11 (GitHub mirror)

Property	Value
Role	Alternative geo-11 distribution
Rendering APIs	DX11
Source	ThreeDeeJay/geo-11
Config file	d3dxdm.ini
Cached core	yes

Places

```
identical to the official build: d3d11.dll, d3dxdm.ini, d3dx.ini
```

A GitHub mirror of geo-11, offered as a separate catalogue entry rather than a silent fallback. The official build is distributed through a blogspot post, which is fragile to scrape; the mirror is a release feed and therefore more reliable, but may lag.

Presenting both as distinct entries means a user always knows which they installed. Silently substituting one for the other would make a version mismatch impossible to diagnose.

Legacy Geo3D

Property	Value
Role	Bundled stable Geo3D
Rendering APIs	DX9, DX10, DX11, DX12
Source	—
Config file	ReShade.ini
Cached core	yes

Places

```
Geo3D.addon64 / Geo3D.addon32
3DToElse.fx
GeoVrExport.addon64   (bundled alongside)
```

A known-good build of Geo3D that ships with the application, labelled “stable (bundled)”. It exists because this specific build is no longer downloadable from its original location, and it is the more reliable of the two for most titles.

The live download is offered alongside it and marked unstable, since it tracks development.

This is the single exception to “nothing is bundled”. It is called out in the interface rather than presented as though it had been downloaded, because a user debugging a problem needs to know which build they have.

SuperVRExport add-on

Property	Value
Role	VR export add-on
Rendering APIs	not API-specific
Source	BerZerker96/Super-VRExport-Addon
Config file	—
Cached core	yes

Places

```
SuperVrExport.addon64 / SuperVrExport.addon32
```

A ReShade add-on that hands SuperDepth3D's stereo frame to a VR viewer at full resolution.

Without it, a headset viewer must capture the game window, and a side-by-side frame in a window of normal width gives each eye half the horizontal pixels. The add-on avoids that entirely.

Pulled in automatically when a VR output format is selected — asserted by test, because the failure mode without it is a black headset view with no obvious cause.

GeoVRExport add-on

Property	Value
Role	VR export add-on
Rendering APIs	not API-specific
Source	BerZerker96/Super-VRExport-Addon
Config file	—
Cached core	yes

Places

```
GeoVrExport.addon64 / GeoVrExport.addon32
```

The Geo3D equivalent, sharing the 3DToElse output. Same rationale, same automatic inclusion.

On a DirectX 9 title it can additionally install a native `geod3d9.dll` proxy for full-resolution VR, which the interface offers as an explicit optional step rather than doing silently.

Osiris VR Viewer

Property	Value
Role	VR viewer
Rendering APIs	not API-specific
Source	BerZerker96/Osiris-Vr-Viewer
Config file	—
Cached core	yes

Places

```
standalone application — not installed into a game folder
```

Displays the side-by-side output in a headset. An alternative to KatangaVR, from the same author as the BerZerker 6DOF hub.

Managed as a cached core so it can be downloaded and updated through the application, but it runs on its own rather than being placed beside a game.

The catalogue link was wrong for a period, pointing at a repository name that no longer existed. It is now `BerZerker96/Osiris-Vr-Viewer`. Worth noting because a broken link in the Mods tab is invisible until someone clicks it.

BepInEx 5 (Unity loader)

Property	Value
Role	Unity mod loader
Rendering APIs	not API-specific
Source	github.com/BepInEx/BepInEx/releases
Config file	—
Cached core	yes

Places

```
winhttp.dll
BepInEx\core\
BepInEx\plugins\      where mod DLLs go
BepInEx\config\       where mod configs go
```

The framework itsloopyo’s head-tracking mods run inside on Unity titles. Installed automatically when such a mod is chosen.

It claims `winhttp.dll`, which is outside the DirectX proxy set, so it does not usually contend with a stereo driver — one of the reasons Unity games are among the easiest to set up with both 3D and 6DOF.

Mod configuration lives in `BepInEx\config\` as INI-like .cfg files, which the configuration editor reads directly.

Ultimate ASI Loader

Property	Value
Role	Generic .asi loader
Rendering APIs	not API-specific
Source	github.com/ThirteenAG/Ultimate-ASI-Loader/releases
Config file	—
Cached core	yes

Places

```
dinput8.dll  (or another proxy name)
*.asi files are loaded from the game folder
```

Loads `.asi` plugins, the format most BerZerker hub mods use on non-Unity games.

In practice the hub mods bundle their own loader, so the application usually does not install this one. That matters: two ASI loaders in one folder is a reliable way to get nothing loading at all, so the mod is checked for a bundled loader first.

It claims `dinput8.dll`, the same slot 3DVision4All prefers. This is the most common rehoming case in practice — see section 18.

REFramework (RE Engine)

Property	Value
Role	RE Engine mod framework
Rendering APIs	not API-specific
Source	github.com/praydog/REFramework-nightly/releases
Config file	—
Cached core	yes

Places

```
dinput8.dll
reframework\
```

The framework mods for Resident Evil and related RE Engine titles run inside. Installed when a chosen mod requires it.

RE Engine games are flat-root, so placement is straightforward, but they are DX12 and therefore outside geo-11's range — SuperDepth3D or Geo3D are the stereo options there.

6DOF Hub (BerZerker96)

Property	Value
Role	6DOF head tracking — BerZerker hub
Rendering APIs	not API-specific
Source	BerZerker96/6DOF-Head-Tracking-Mods-Hub
Config file	—
Cached core	yes

Places

```
plain builds:    HeadTracking.asi + HeadTracking.ini  (+ bundled ASI loader)
combined builds: version.dll + “3D-6DOF config.ini”
```

Around fifty per-game mods published as separate releases in one repository. Downloaded per game rather than cached as a shared core, because each release is a different mod.

A subset are combined 3D + 6DOF builds carrying their own DX11 stereo implementation, described in detail in section 12. Those need no stereo driver at all and should not be paired with one.

The manual picker can install any release onto any game, which is the correct behaviour for a catalogue that cannot possibly know every folder name a user might have.

Release titles and git tags frequently differ. ACOrigins is tagged `1`, ACSyndicate `s1`, ACUnity `u1`. Resolution tries both.

Head-Tracking (itsloopyo)

Property	Value
Role	6DOF head tracking — itsloopyo
Rendering APIs	not API-specific
Source	—
Config file	—
Cached core	no — downloaded per game

Places

```
BepInEx\plugins\HeadTracking.dll
BepInEx\config\*.headtracking.cfg
```

Around thirty mods, one repository per game, roughly half publishing only pre-releases.

Mostly Unity titles, so BepInEx is the usual host. Configuration is a BepInEx .cfg file with sections for the UDP connection, sensitivity and tracking behaviour.

Like the hub, downloaded per game and installable onto any game through the manual picker.

Two source-specific traps. GitHub's `/releases/latest` excludes pre-releases and 404s when a repository has only those, which hid fourteen of thirty mods. And pre-release builds ship only `-installer.zip` where released builds ship `-nexus.zip`, so an asset filter excluding installers excluded the only asset present.

dgVoodoo2 (DX8/9/10 → DX11)

Property	Value
Role	Legacy DirectX and Glide wrapper
Rendering APIs	not API-specific
Source	dege-diosg/dgVoodoo2
Config file	dgVoodoo.conf
Cached core	yes

Places

```
D3D8.dll  D3D9.dll  D3DImm.dll  DDraw.dll
dgVoodooCpl.exe
dgVoodoo.conf
```

Covered in full in section 13. Included here for completeness: it is a cached core like any other, fetched from Dege's own release feed, and it has its own installer rather than going through the generic placement path.

The complete MS set for the game's bitness is installed, not only the DLL matching the detected API, because a game reported as DX9 often also creates a DirectDraw or D3D8 device.

Appendix H. Glossary

Term	Meaning
Proxy DLL	A library named after a system DLL so the game loads it instead, letting it intercept calls
Proxy slot	A specific proxy filename. Only one file can hold it, so only one mod can use it
Rehoming	Renaming an installed mod to a different proxy slot so another mod can use the first
Core	A cached mod payload, downloaded once and linked into many games
Manifest	The per-game record of exactly what this application placed there
Depth-buffer 3D	Reconstructing a second eye view from the depth buffer. Works everywhere; imperfect on transparency
Geometric 3D	Rendering the scene twice with a shifted camera. Correct geometry; needs per-game fixes
Shader fix	A community patch correcting effects that break under a shifted camera
Separation	The distance between the two virtual eyes — how strong the 3D effect is
Convergence	Where the two views coincide — what appears at screen depth
SBS	Side-by-side: both eye views in one frame, left and right
TAB	Top-and-bottom: both eye views stacked vertically
6DOF	Six degrees of freedom — positional and rotational camera control from head tracking
ASI	A plugin format loaded by an ASI loader, used by many game mods
Add-on	A ReShade plugin, as distinct from a shader. Requires the add-on build of ReShade
Shipping build	An Unreal release binary, named *-Win64-Shipping.exe
Combined build	A 6DOF mod that also includes its own stereo implementation

End of document. Build 86fc41b8. Generated 2026-08-09.